Exp 4:

server:

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<arpa/inet.h>
#include<unistd.h>
#include<time.h>

int main()
{
        int server_sock;
        char buffer[1024],now[100];
        time_t t;
        struct sockaddr_in serverAddr, clientAddr;
        server_sock = socket(AF_INET, SOCK_DGRAM, 0);
        serverAddr.sin_family = AF_INET;
        serverAddr.sin_port = htons(7564);
        serverAddr.sin_addr.s_addr = INADDR_ANY;
        bind(server_sock, (struct sockaddr *)&serverAddr, sizeof(serverAddr));
        printf("UDP Server Running...\n");
        socklen_t len=sizeof(clientAddr);
        while(1)
        {
                recvfrom(server_sock, buffer, sizeof(buffer), 0,(struct sockaddr
*)&clientAddr,&len);
        time(&t);
        strcpy(now,ctime(&t));
        sendto(server_sock,now,sizeof(now),0,(struct sockaddr*)&clientAddr,len);
      }
        close(server_sock);
        return 0;
}
```

client:

```
#include <stdio.h>
#include <string.h>
#include <arpa/inet.h>
#include <unistd.h>
```

```
int main()
{
int sockfd;
char buffer[1024];

struct sockaddr_in serverAddr;

sockfd = socket(AF_INET, SOCK_DGRAM, 0);

serverAddr.sin_family = AF_INET;
serverAddr.sin_port = htons(7564);
serverAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

strcpy(buffer,"TIME REQUEST");
sendto(sockfd,buffer,strlen(buffer)+1,0,(struct sockaddr*)&serverAddr,sizeof(serverAddr));
recvfrom(sockfd,buffer,sizeof(buffer),0,NULL,NULL);

printf("\nCurrent System Time:\n%s\n",buffer);

close(sockfd);

return 0;
}
```

output:
//server
UDP Server Running...
//client
Current System Time:
Wed Sep 9 09:53:56 2026

Exp 5:

server:

```
#include<stdio.h>
#include<stdlib.h>
```